3DWEBSOFT INC. FOUNDATION

FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Python Mini Project



FOUNDATION

1. What is a Python Project?

A **Python project** is a small or large program created to solve a real-life problem.

Instead of learning Python topics separately, a project helps us **combine different topics together**.

For example:

Variables

↓

Input

↓

Operators

↓

Conditions

↓

Lists

↓

Functions

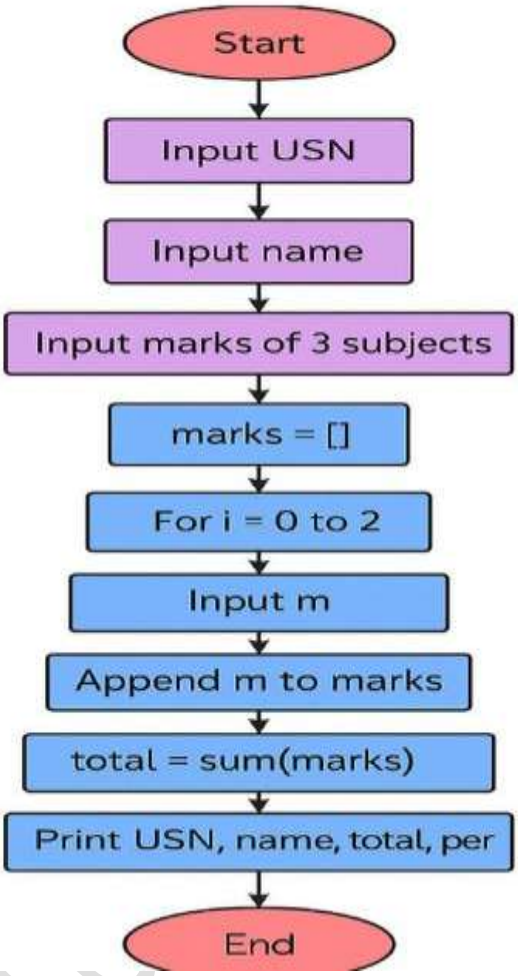
↓

Exception Handling

↓

Python Project

3DWEBSOFT FOUNDATION


Start
Input USN
Input name
Input marks of 3 subjects
marks = []
For i = 0 to 2
Input m
Append m to marks
total = sum(marks)
Print USN, name, total, per
End


REPORT CARD
NAME:
CLASS:
DATE:
A+

2. Learning Objectives

By the end of this lesson, students will be able to:

- Understand what a mini project is.
- Plan a simple Python program.
- Use input().
- Use variables.
- Perform calculations.
- Use if, elif, and else.
- Use lists.
- Create functions.
- Use loops.
- Handle simple errors.
- Build a complete Python application.

3. Why Do We Build Projects?

Learning programming is like learning to play cricket.

Knowing how to hold a bat is not enough.

You need to **play a game**.

Similarly:

Learning Python concepts

↓

Practice small programs

↓

Combine concepts

↓

3DWEBSOFT FOUNDATION

Build a project

↓

Become a better programmer

4. What Will We Build?

Today we will create:

Student Report Card

The program will:

1. Ask the student's name.
2. Ask the student's class.
3. Ask for marks.
4. Calculate total marks.
5. Calculate average.
6. Find the grade.
7. Display the result.
8. Handle incorrect input.

Example

```
==============================
    STUDENT REPORT CARD
==============================

Name: Aisha

Class: 8

Math: 90
```

3DWEBSOFT FOUNDATION

Science: 85

English: 88

Total: 263

Average: 87.67

Grade: A

5. Project Planning

Before writing code, programmers make a plan.

Step 1 – Identify the Problem

Problem: Students need a simple way to calculate their marks and grade.

Step 2 – Decide Inputs

We need:

- Student name
- Class
- Math marks
- Science marks
- English marks

Step 3 – Process

Calculate:

Total = Math + Science + English

Average = Total / 3

Step 4 – Output

Display:

- Name

3DWEBSOFT FOUNDATION

- Class
- Total
- Average
- Grade

Project Flowchart

START

↓

Enter Student Name

↓

Enter Class

↓

Enter 3 Marks

↓

Calculate Total

↓

Calculate Average

↓

Find Grade

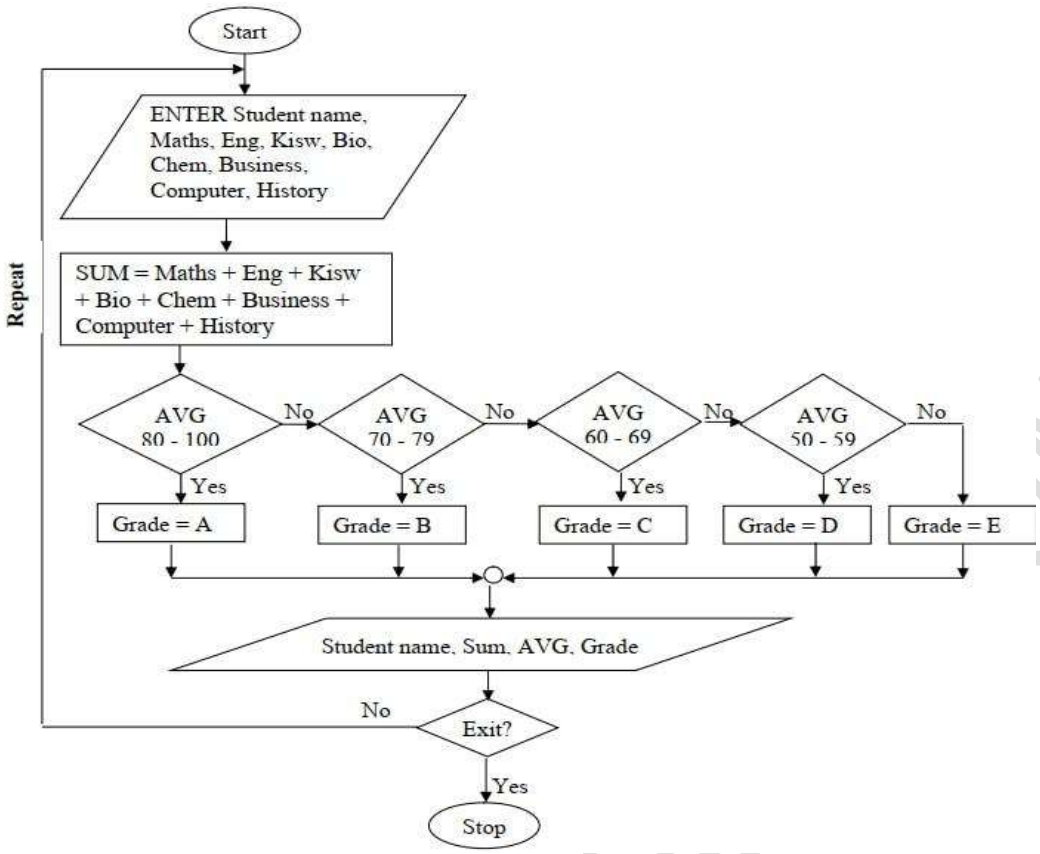
↓

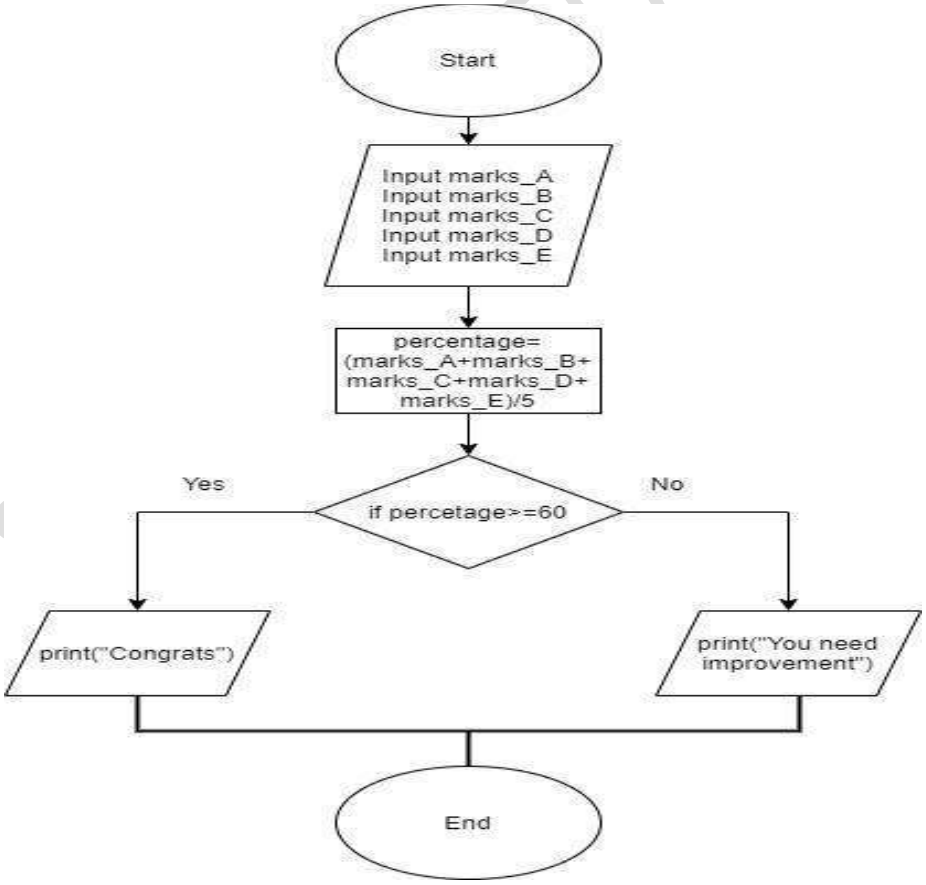
Display Report Card

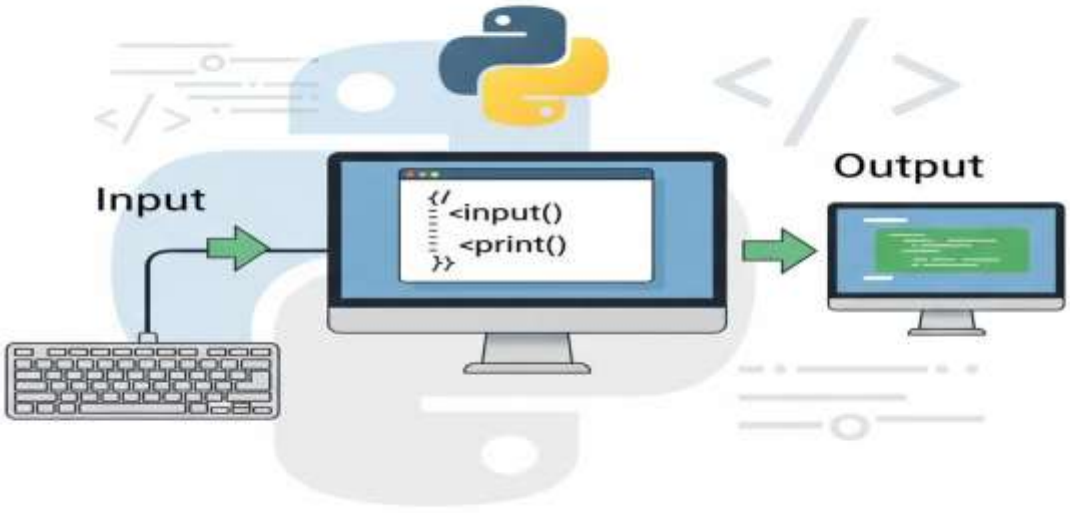
↓

END

3DWEBSOFT FOUNDATION


Start
ENTER Student name, Maths, Eng, Kisw, Bio, Chem, Business, Computer, History
Repeat
SUM = Maths + Eng + Kisw + Bio + Chem + Business + Computer + History
AVG 80 - 100
No
AVG 70 - 79
No
AVG 60 - 69
No
AVG 50 - 59
No
Yes
Yes
Yes
Yes
Grade = A
Grade = B
Grade = C
Grade = D
Grade = E
Student name, Sum, AVG, Grade
No
Exit?
Yes
Stop


Start
Input marks_A
Input marks_B
Input marks_C
Input marks_D
Input marks_E
percentage= (marks_A+marks_B+ marks_C+marks_D+ marks_E)/5
Yes
if percetage>=60
No
print("Congrats")
print("You need improvement")
End



6. Python Concepts Used

This project combines concepts students have already learned.

Python Concept	How We Use It
Variables	Store student information
input()	Get information
int()	Convert marks to numbers
Operators	Calculate total and average
if	Check conditions
elif	Check multiple grades
else	Handle remaining cases
Lists	Store marks

3DWEBSOFT FOU

Python Concept	How We Use It
Functions	Organize code
Loops	Process marks
Exception Handling	Handle invalid input

7. Step 1 – Get Student Name

```
name = input("Enter student name: ")


print("Welcome", name)
```

Example

Enter student name: Aisha

Welcome Aisha

8. Step 2 – Get Class

```
student_class = input("Enter class: ")


print("Class:", student_class)
```

9. Step 3 – Get Marks

```
math = int(input("Enter Math marks: "))
science = int(input("Enter Science marks: "))
english = int(input("Enter English marks: "))
```

Example:

Math: 90

Science: 85

3DWEBSOFT FOUNDATION

English: 88

10. Step 4 – Calculate Total

total = math + science + english

print("Total:", total)

Example:

Total: 263

11. Step 5 – Calculate Average

average = total / 3

print("Average:", average)

Example:

Average: 87.66666666666667

We can make it easier to read using:

print("Average:", round(average, 2))

Output:

Average: 87.67

12. Step 6 – Find the Grade

We can use conditional statements.

if average >= 90:

    grade = "A+"

3DWEBSOFT FOUNDATION

```
elif average >= 80:
    grade = "A"

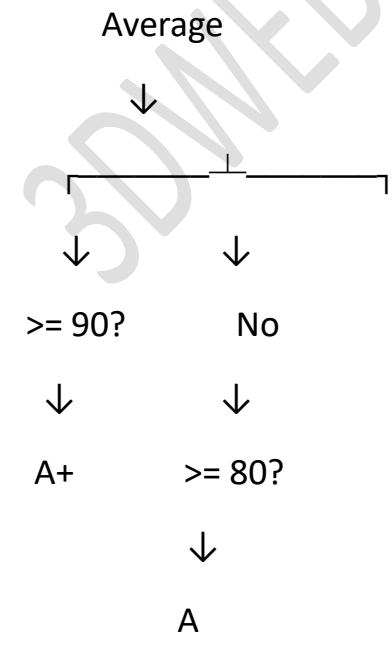
elif average >= 70:
    grade = "B"

elif average >= 60:
    grade = "C"

elif average >= 40:
    grade = "D"

else:
    grade = "F"
```

Grade Decision Diagram



3DWEBSOFT FOUNDATION

↓

>= 70?

↓

B

↓

>= 60?

↓

C

↓

>= 40?

↓

D

↓

F

13. Step 7 – Display Report Card

```
print("============================")
print("      STUDENT REPORT CARD")
print("============================")

print("Name:", name)
print("Class:", student_class)

print("Math:", math)
print("Science:", science)
```

3DWEBSOFT FOUNDATION

```
print("English:", english)


print("Total:", total)
print("Average:", round(average, 2))
print("Grade:", grade)

print("==============================")
```

14. Complete Mini Project – Version 1

```
print("==============================")
print("      STUDENT REPORT CARD")
print("==============================")

name = input("Enter student name: ")
student_class = input("Enter class: ")

math = int(input("Enter Math marks: "))
science = int(input("Enter Science marks: "))
english = int(input("Enter English marks: "))

total = math + science + english
average = total / 3

if average >= 90:
    grade = "A+"
```

3DWEBSOFT FOUNDATION

```
elif average >= 80:
    grade = "A"
elif average >= 70:
    grade = "B"
elif average >= 60:
    grade = "C"
elif average >= 40:
    grade = "D"
else:
    grade = "F"

print()
print("==============================")
print("      STUDENT REPORT CARD")
print("==============================")

print("Name:", name)
print("Class:", student_class)
print("Math:", math)
print("Science:", science)
print("English:", english)
print("Total:", total)
print("Average:", round(average, 2))
print("Grade:", grade)
```

3DWEBSOFT FOUNDATION

```
print("==============================")
```

15. Sample Run

Input

Enter student name: Aisha

Enter class: 8

Enter Math marks: 90

Enter Science marks: 85

Enter English marks: 88

Output

```
==============================
      STUDENT REPORT CARD
==============================

Name: Aisha
Class: 8
Math: 90
Science: 85
English: 88
Total: 263
Average: 87.67
Grade: A

==============================
```

You have created your first complete Python project.

3DWEBSOFT FOUNDATION

16. Improving the Project with Lists

Instead of creating separate variables for every subject, we can use a **List**.

```
marks = [90, 85, 88]


print(marks)
```

Output:

```
[90, 85, 88]
```

Calculate Total Using a List

Python has the sum() function.

```
marks = [90, 85, 88]


total = sum(marks)


print("Total:", total)
```

Output:

```
Total: 263
```

Calculate Average

```
marks = [90, 85, 88]


total = sum(marks)

average = total / len(marks)


print("Average:", average)
```

3DWEBSOFT FOUNDATION

17. Using a Loop

We can ask for marks using a loop.

```
marks = []


for i in range(3):
    mark = int(input("Enter marks: "))
    marks.append(mark)


print(marks)
```

Example:

```
Enter marks: 90
Enter marks: 85
Enter marks: 88


[90, 85, 88]
```

18. Why Use a Loop?

Imagine there are 10 subjects.

Without a loop:

```
mark1 = int(input())
mark2 = int(input())
mark3 = int(input())
...
```

This becomes long.

With a loop:

3DWEBSOFT FOUNDATION

```
for i in range(10):
    mark = int(input())
```

Much easier!

19. Add Exception Handling

What if a student enters:

ninety

instead of:

90

The program may produce a ValueError.

We can safely handle it.

```
try:
    mark = int(input("Enter marks: "))
    print("Marks:", mark)

except ValueError:
    print("Please enter numbers only.")
```

20. Improved Project

Now let's combine:

- Input
- List
- Loop
- Operators
- Conditions

3DWEBSOFT FOUNDATION

- Exception Handling

```
print("================================")
print("       STUDENT REPORT CARD")
print("================================")

name = input("Enter student name: ")
student_class = input("Enter class: ")

marks = []

try:
    for i in range(3):
        mark = int(input("Enter marks: "))
        marks.append(mark)

    total = sum(marks)
    average = total / len(marks)

    if average >= 90:
        grade = "A+"
    elif average >= 80:
        grade = "A"
    elif average >= 70:
        grade = "B"
    elif average >= 60:
```

3DWEBSOFT FOUNDATION

```
        grade = "C"
    elif average >= 40:
        grade = "D"
    else:
        grade = "F"

    print()
    print("================================")
    print("      STUDENT REPORT CARD")
    print("================================")

    print("Name:", name)
    print("Class:", student_class)
    print("Marks:", marks)
    print("Total:", total)
    print("Average:", round(average, 2))
    print("Grade:", grade)

except ValueError:
    print("⚠️ Please enter marks using numbers only.")

finally:
    print("Thank you for using the program!")
```

21. Adding Functions

3DWEBSOFT FOUNDATION

A **function** is a reusable block of code.

We can create:

```
def calculate_grade(average):
    if average >= 90:
        return "A+"
    elif average >= 80:
        return "A"
    elif average >= 70:
        return "B"
    elif average >= 60:
        return "C"
    elif average >= 40:
        return "D"
    else:
        return "F"
```

Then:

```
grade = calculate_grade(average)
```

This makes the program easier to understand.

22. Final Project Version

```
def calculate_grade(average):
    if average >= 90:
        return "A+"
    elif average >= 80:
        return "A"
```

3DWEBSOFT FOUNDATION

```
    elif average >= 70:
        return "B"
    elif average >= 60:
        return "C"
    elif average >= 40:
        return "D"
    else:
        return "F"



print("================================")
print("      STUDENT REPORT CARD")
print("================================")

name = input("Enter student name: ")
student_class = input("Enter class: ")

marks = []

try:
    for i in range(3):
        mark = int(input("Enter marks: "))

        if mark < 0 or mark > 100:
            print("Marks must be between 0 and 100.")
```

3DWEBSOFT FOUNDATION

```
        raise ValueError

    marks.append(mark)

total = sum(marks)
average = total / len(marks)

grade = calculate_grade(average)

print()
print("================================")
print("      STUDENT REPORT CARD")
print("================================")

print("Name:", name)
print("Class:", student_class)
print("Marks:", marks)
print("Total:", total)
print("Average:", round(average, 2))
print("Grade:", grade)

if grade == "A+":
    print("Excellent performance! ")
elif grade == "A":
    print("Very good! ")
```

3DWEBSOFT FOUNDATION

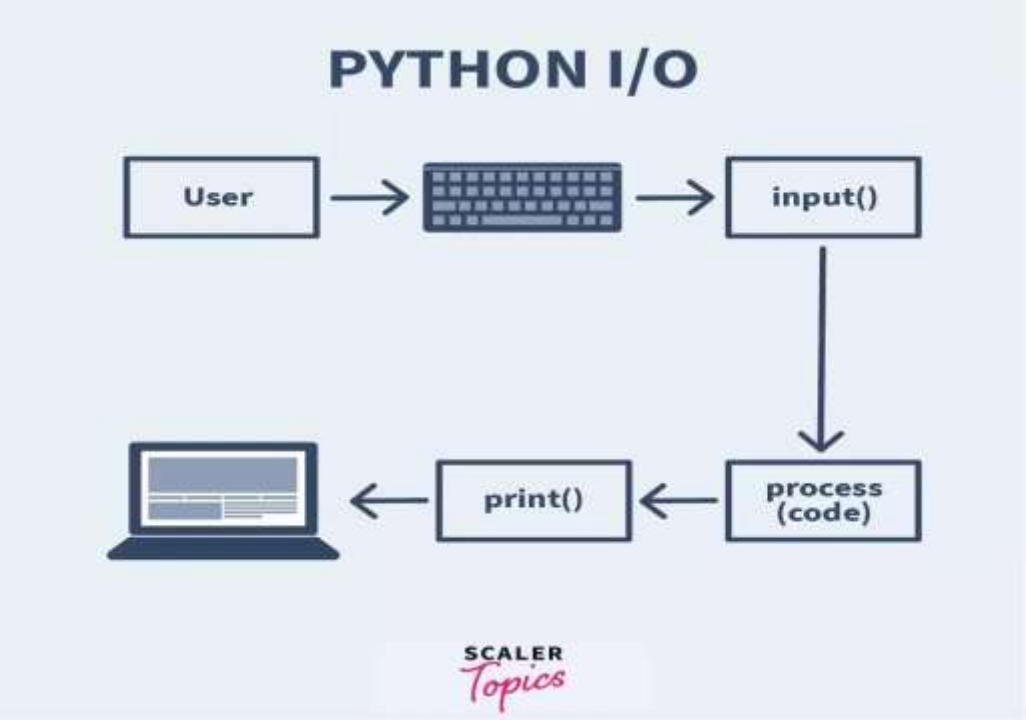
```
    elif grade == "B":
        print("Good work! ")
    else:
        print("Keep practicing! ")

except ValueError:
    print(" Please enter valid marks.")

finally:
    print()
    print("Thank you for using the Student Report Card!")
```

23. Project Architecture

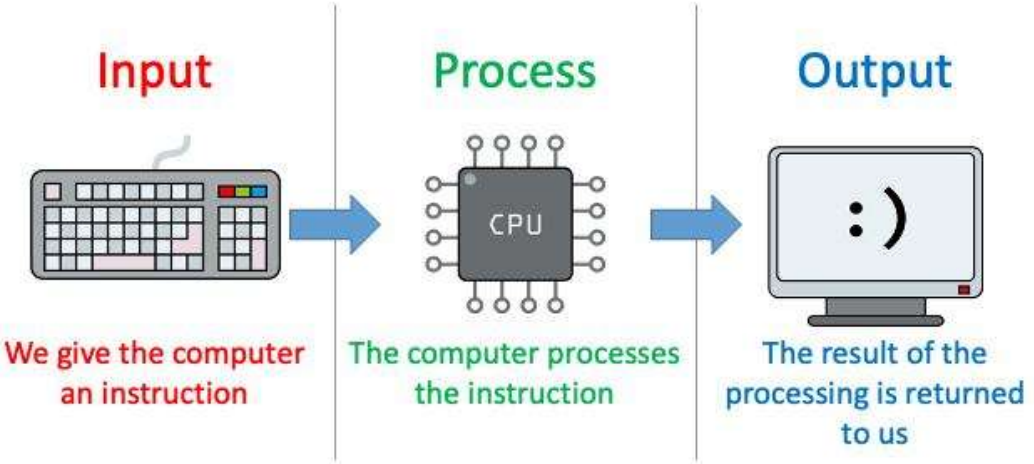
Students can understand the program like this:

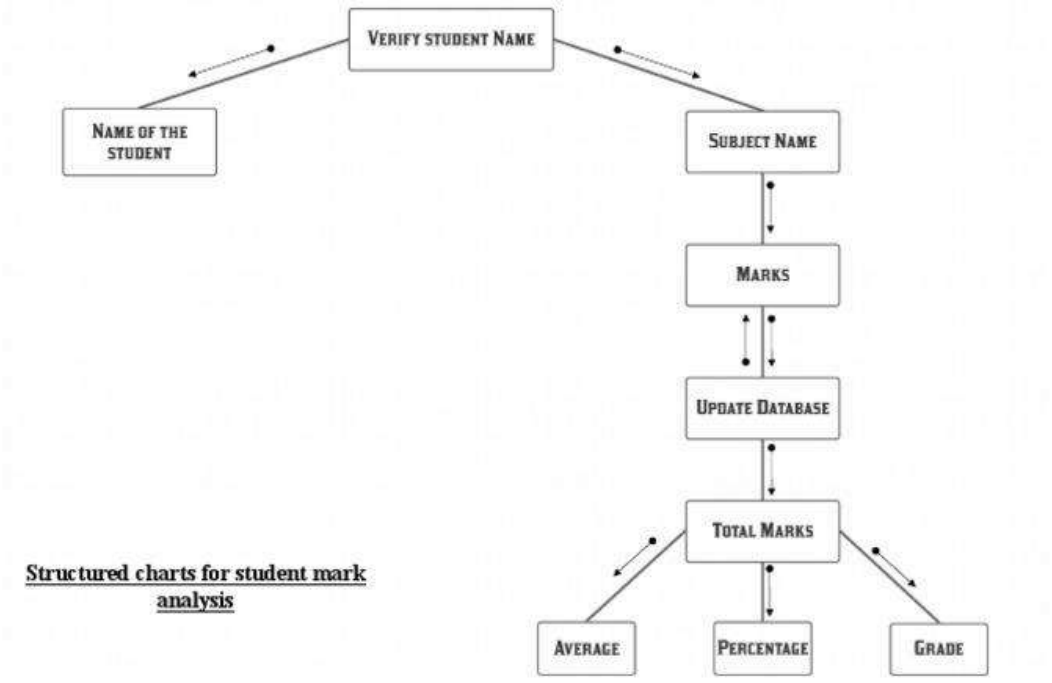


ET FOUNDATION

Input – Process – Output

Computers also work like this

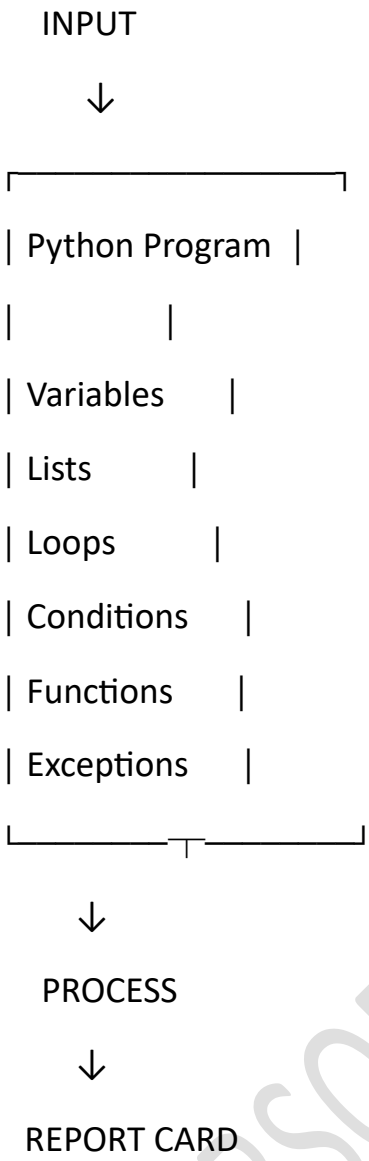




Structured charts for student mark analysis

STUDENT

↓



3DWEBSOFT FOUNDATION

24. Student Challenge

Modify the project to include:

Challenge 1

Add **Telugu marks**.

Challenge 2

Add **Hindi marks**.

Challenge 3

Display the **highest mark**.

highest = max(marks)

print("Highest:", highest)

Challenge 4

Display the **lowest mark**.

lowest = min(marks)

print("Lowest:", lowest)

Challenge 5

Display number of subjects.

print("Subjects:", len(marks))

25. Mini Project Ideas for Students

After completing the Report Card project, students can build other beginner projects.

Project 1 – Calculator

Operations:

+

-

*

/

Project 2 – Number Guessing Game

Computer chooses a number and the student guesses it.

Project 3 – Quiz Game

Ask questions and calculate score.

Project 4 – To-Do List

3DWEBSOFT FOUNDATION

Add and display tasks.

Project 5 – Simple Shopping List

Add items and display the list.

Project 6 – Temperature Converter

Convert:

Celsius → Fahrenheit

Fahrenheit → Celsius

Project 7 – Attendance Calculator

Calculate attendance percentage.

Project 8 – Simple Password Checker

Check whether a password meets simple rules.

26. Classroom Practical Activity

Divide students into small groups.

Team A

Build a **Calculator**.

Team B

Build a **Quiz Game**.

Team C

Build a **To-Do List**.

Team D

Build a **Student Report Card**.

Each team should explain:

1. What problem does the project solve?
2. What inputs are required?

3DWEBSOFT FOUNDATION

3. What Python concepts were used?
4. What output is produced?

27. Project Documentation

Students should prepare a simple project record.

Project Name

Student Report Card

Problem

Students need a simple program to calculate marks and grades.

Inputs

- Name
- Class
- Subject marks

Processing

- Total
- Average
- Grade

Output

- Student report card

Python Concepts

- Variables
- Input
- Lists
- Loops
- Conditions

3DWEBSOFT FOUNDATION

- Functions
- Exception Handling

Quiz

1. What is a Python project?

A) A computer part
B) A program created to solve a problem
C) A keyboard
D) A folder

✅ **Answer: B**

2. Which function gets input from the user?

A) print()

B) input()

C) get()

D) read()

✅ **Answer: B**

3. Which function finds the total of numbers in a list?

A) total()

B) sum()

C) add()

D) plus()

✅ **Answer: B**

3DWEBSOFT FOUNDATION

4. Which function finds the number of items?

A) count()

B) size()

C) len()

D) number()

✅ **Answer: C**

5. Which keyword creates a function?

A) function

B) def

C) fun

D) create

✅ **Answer: B**

6. Which block handles errors?

A) try/except

B) if/else

C) for/while

D) print/input

✅ **Answer: A**

Homework

Build Your Own Mini Project

Choose **one**:

3DWEBSOFT FOUNDATION

☐ Calculator

☐ Quiz Game

☐ Number Guessing Game

☐ To-Do List

☐ Shopping List

☐ Student Report Card

Your project must contain at least:

- Variables
- input()
- print()
- if/elif/else
- One loop OR list
- Exception handling

30. Project Evaluation

Skill	**Marks**
Project idea	10
Correct Python code	20
Input & Output	10
Conditions	10
List/Loop	10
Exception Handling	10
Creativity	10
Presentation	10

3DWEBSOFT FOUNDATION

Skill	Marks
Explanation	10
Total	**100**

31. Project Presentation Questions

Students should be able to answer:

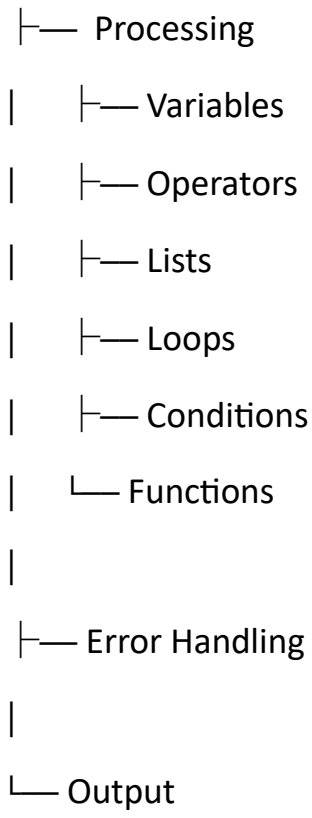
1. What is your project name?
2. What problem does it solve?
3. What inputs does your program take?
4. Which Python concepts did you use?
5. Why did you use a list?
6. Why did you use a loop?
7. Why did you use a function?
8. What happens if the user enters invalid data?
9. What was the most difficult part?
10. What would you improve in the future?

32. Final Revision

PYTHON MINI PROJECT

```
|
├— Problem
|
├— Input
|
```

3DWEBSOFT FOUNDATION



3DWEBSOFT FOUNDATION